



UNSW
SYDNEY

MATH2601 – Higher Linear Algebra

Topic 6 – Eigenvalues and eigenvectors

Lecture 6.2 – Normal operators and spectral theorems

Lecturer: Dr Sean Gardiner – sean.gardiner@unsw.edu.au

Normal maps and matrices

Definition. Given an inner product space V and $T \in \mathcal{L}(V, V)$, the mapping T is called **normal** if and only if $T^* \circ T = T \circ T^*$. Likewise, a matrix $A \in M_{n,n}(\mathbb{F})$ is called normal if and only if $AA^* = A^*A$.

Clearly all self-adjoint (Hermitian) maps/matrices are normal since if $T^* = T$ then certainly $T^* \circ T = T \circ T^*$, and all unitary maps/matrices are normal since if $T^* = T^{-1}$ then $T^* \circ T = \text{id} = T \circ T^*$.

An example of a matrix that is normal but not self-adjoint nor unitary is

$A = \begin{pmatrix} 1+i & 2i \\ 2 & 2+2i \end{pmatrix}$. Notice that $A^* = \begin{pmatrix} 1-i & 2 \\ -2i & 2-2i \end{pmatrix}$ and that

$$AA^* = \begin{pmatrix} 6 & 6+6i \\ 6-6i & 12 \end{pmatrix} = A^*A.$$

Properties of normal maps

Lemma. A map $T \in \mathcal{L}(V, V)$ is **normal** if and only if $\|T(\mathbf{v})\| = \|T^*(\mathbf{v})\|$ for all $\mathbf{v} \in V$.

Proof. If T is normal, then $\|T(\mathbf{v})\|^2 = \langle T(\mathbf{v}), T(\mathbf{v}) \rangle = \langle T^* \circ T(\mathbf{v}), \mathbf{v} \rangle$ is equal to $\langle T \circ T^*(\mathbf{v}), \mathbf{v} \rangle = \langle (T^*)^* \circ T^*(\mathbf{v}), \mathbf{v} \rangle = \langle T^*(\mathbf{v}), T^*(\mathbf{v}) \rangle = \|T^*(\mathbf{v})\|^2$.

For the reverse direction, if $\|T(\mathbf{v})\| = \|T^*(\mathbf{v})\|$ for all $\mathbf{v} \in V$, then $0 = \langle T(\mathbf{v}), T(\mathbf{v}) \rangle - \langle T^*(\mathbf{v}), T^*(\mathbf{v}) \rangle = \langle (T^* \circ T - T \circ T^*)(\mathbf{v}), \mathbf{v} \rangle$ for all $\mathbf{v} \in V$, so $T^* \circ T = T \circ T^*$ meaning T is normal.

Lemma. Given $T \in \mathcal{L}(V, V)$ is normal, if λ is an eigenvalue for T then $\bar{\lambda}$ is an eigenvalue for T^* , and $E_\lambda(T) = E_{\bar{\lambda}}(T^*)$.

Proof. If $\mathbf{v} \in E_\lambda(T)$, then $T(\mathbf{v}) = \lambda\mathbf{v}$ implying $0 = \|(T - \lambda \text{id})(\mathbf{v})\|$, so by the first lemma $0 = \|(T - \lambda \text{id})^*(\mathbf{v})\| = \|(T^* - \bar{\lambda} \text{id})(\mathbf{v})\|$. This implies $T^*(\mathbf{v}) = \bar{\lambda}\mathbf{v}$, meaning $\bar{\lambda}$ is an eigenvalue of T^* and $\mathbf{v} \in E_{\bar{\lambda}}(T^*)$. This shows that $E_\lambda(T) \subseteq E_{\bar{\lambda}}(T^*)$, and the reverse containment follows similarly.

Lemma. Given $T \in \mathcal{L}(V, V)$ is normal, if λ and μ are distinct eigenvalues for T , then E_λ and E_μ are orthogonal.

Proof. Let $\mathbf{u} \in E_\lambda$ and $\mathbf{v} \in E_\mu$ for distinct eigenvalues λ, μ . Then $\lambda \langle \mathbf{v}, \mathbf{u} \rangle = \langle \mathbf{v}, \lambda \mathbf{u} \rangle = \langle \mathbf{v}, T(\mathbf{u}) \rangle = \langle T^*(\mathbf{v}), \mathbf{u} \rangle = \langle \bar{\mu} \mathbf{v}, \mathbf{u} \rangle = \bar{\mu} \langle \mathbf{v}, \mathbf{u} \rangle$. So $\langle \mathbf{v}, \mathbf{u} \rangle = 0$, meaning E_λ and E_μ are orthogonal.

Multiplicities for normal maps

Theorem. If $T \in \mathcal{L}(V, V)$ is normal, then for each eigenvalue λ of T , its algebraic multiplicity and geometric multiplicity are equal.

Proof. Consider the decomposition of V into orthogonal complements $V = E_\lambda \oplus E_\lambda^\perp$. If $\dim(E_\lambda) = k$, we can set up bases $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ for E_λ and $\{\mathbf{u}_{k+1}, \dots, \mathbf{u}_n\}$ for $E_\lambda^\perp$ whose union B forms a basis for V .

Let $\mathbf{v} \in E_\lambda$. Notice $T(T^*(\mathbf{v})) = T^*(T(\mathbf{v})) = T^*(\lambda\mathbf{v}) = \lambda T^*(\mathbf{v})$, so $T^*(\mathbf{v}) \in E_\lambda$. So E_λ is T^* -invariant, meaning $T^*(E_\lambda) \leq E_\lambda$.

Now suppose $\mathbf{w} \in E_\lambda^\perp$. Notice $\langle \mathbf{v}, T(\mathbf{w}) \rangle = \langle T^*(\mathbf{v}), \mathbf{w} \rangle = 0$ since $T^*(\mathbf{v}) \in E_\lambda$. So $E_\lambda^\perp$ is T -invariant, meaning $T(E_\lambda^\perp) \leq E_\lambda^\perp$.

Hence $[T]_B^B = \begin{pmatrix} \lambda I_k & \mathbf{0} \\ \mathbf{0} & X \end{pmatrix}$ for some $X \in M_{n-k, n-k}(\mathbb{F})$.

So $p_T(t) = (t - \lambda)^k p_X(t)$. Notice that if λ were an eigenvalue for X , this would imply there is a corresponding eigenvector $\mathbf{v}$ which is both in E_λ by definition and in $E_\lambda^\perp$ by the fact X is the transformation matrix of T restricted to $E_\lambda^\perp$. But orthogonal complements have trivial intersection and eigenvectors must be nonzero, so this is a contradiction. Thus there are no factors of $(t - \lambda)$ in $p_X(t)$ and so the algebraic multiplicity of λ is k , the same as the geometric multiplicity.

Spectral theorem for normal operators

A **spectral theorem** for a particular class of operators is a result that determines when they are diagonalisable.

Theorem. (Spectral theorem for normal operators)

Given an inner product space V over $\mathbb{C}$ with finite dimension n , any map $T \in \mathcal{L}(V, V)$ or matrix $A \in M_{n,n}(\mathbb{C})$ is **normal** if and only if it is **unitarily diagonalisable**, meaning:

- For $A \in M_{n,n}(\mathbb{C})$, there is some **unitary** $P \in \text{GL}(n, \mathbb{C})$ for which $P^{-1}AP = P^*AP$ is diagonal.
- For $T \in \mathcal{L}(V, V)$, there is an **orthonormal** basis B of V made up of eigenvectors of T (so $[T]_B^B = PDP^{-1}$ for some diagonal $D \in M_{n,n}(\mathbb{C})$ and **unitary** $P \in \text{GL}(n, \mathbb{C})$).

Proof. We already know that for any diagonalisable map T , we have $V = E_{\lambda_1} \oplus \cdots \oplus E_{\lambda_k}$ where T has k distinct eigenvalues λ_i . We have also shown that each eigenspace is orthogonal whenever T is normal. So if we select linearly independent eigenvectors from the eigenspaces as usual, scaling them to have length 1, the resulting set of eigenvectors will be an orthonormal basis of V , and likewise the corresponding matrices P will be unitary.

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Proof (continued). For the reverse direction, suppose $[T]_B^B = A = PDP^{-1} = PDP^*$ for diagonal D and unitary P . Then

$$A^*A = (PDP^*)^*(PDP^*) = (PD^*P^*)(PDP^*) = P\overline{D}DP^*,$$

where $D^* = \overline{D}$ since D is diagonal. But noting that $\overline{D}D = D\overline{D}$, we have $A^*A = P\overline{D}DP^* = (PDP^*)(PDP^*)^* = AA^*$ by the same argument, so A is normal (as is T).

Example

Example. A unitary diagonalisation for the normal matrix

$$A = \begin{pmatrix} 1+i & 2i \\ 2 & 2+2i \end{pmatrix} \text{ is } D = P^{-1}AP \text{ where } D = \begin{pmatrix} 0 & 0 \\ 0 & 3+3i \end{pmatrix} \text{ and}$$

$$P = \frac{1}{\sqrt{6}} \begin{pmatrix} 2i & \sqrt{2}i \\ -1-i & \sqrt{2} + \sqrt{2}i \end{pmatrix}. \text{ (Check this!)}$$

Corollary. Given a normal matrix $A \in M_{n,n}(\mathbb{C})$ with eigenvalues $\lambda_1, \dots, \lambda_n$ (counting algebraic multiplicities), there exist corresponding unitary mappings $P_i \in \mathcal{L}(\mathbb{C}^n, \mathbb{C}^n)$ for which $P_i^2 = P_i$ for all i , $P_i P_j = \mathbf{0}$ for all $i \neq j$, $\sum_{i=1}^n P_i = \text{id}$, and $\sum_{i=1}^n \lambda_i P_i = T$ where $T(\mathbf{v}) = A\mathbf{v}$ for all $\mathbf{v} \in \mathbb{C}^n$.

Proof. Set $P_i = \text{proj}_{E_{\lambda_i}}$ for each i . Then since the eigenspaces are all orthogonal to each other, the above results follow from properties previously proven about orthogonal projections.

Spectral theorem for self-adjoint operators

Corollary. (Spectral theorem for self-adjoint operators)

Given an inner product space V over $\mathbb{R}$ or $\mathbb{C}$ with finite dimension n , and any self-adjoint map $T \in \mathcal{L}(V, V)$ or matrix $A \in M_{n,n}(\mathbb{F})$, there exists an orthonormal basis for V made up of eigenvectors of T or A (and in the case that $\mathbb{F} = \mathbb{R}$, the eigenvectors are all real).

Proof. This follows immediately from the fact any self-adjoint operator is normal and the spectral theorem for normal operators.

Theorem. Given an inner product space V over $\mathbb{R}$ or $\mathbb{C}$ with finite dimension n , any self-adjoint map $T \in \mathcal{L}(V, V)$ or matrix $A \in M_{n,n}(\mathbb{F})$ has real eigenvalues.

Proof. Suppose $T(\mathbf{v}) = \lambda\mathbf{v}$ for some nonzero $\mathbf{v} \in V$ and $\lambda \in \mathbb{F}$. Then $\lambda\langle\mathbf{v}, \mathbf{v}\rangle = \langle\mathbf{v}, \lambda\mathbf{v}\rangle = \langle\mathbf{v}, T(\mathbf{v})\rangle = \langle T^*(\mathbf{v}), \mathbf{v}\rangle$. Since T is self-adjoint, this is the same as $\langle T(\mathbf{v}), \mathbf{v}\rangle = \langle\lambda\mathbf{v}, \mathbf{v}\rangle = \bar{\lambda}\langle\mathbf{v}, \mathbf{v}\rangle$. Since $\mathbf{v} \neq \mathbf{0}$, this implies that $\lambda = \bar{\lambda}$, meaning each eigenvalue must be real.

Example

Exercise. Unitarily diagonalise the matrix $A = \begin{pmatrix} 3 & -i \\ i & 3 \end{pmatrix}$.

Solution. Since A is self-adjoint, the eigenvalues must be real. Indeed, we have $p_A(t) = (t - 3)^2 + i^2 = (t - 2)(t - 4)$, so the eigenvalues are 2 and 4.

We find $E_2 = \text{span} \left\{ \begin{pmatrix} i \\ 1 \end{pmatrix} \right\}$ and $E_4 = \text{span} \left\{ \begin{pmatrix} i \\ -1 \end{pmatrix} \right\}$. So the two representative eigenvectors are orthogonal, and dividing through by $\sqrt{2}$ makes them orthonormal. Thus $D = P^{-1}AP$ where $D = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$ and

$P = \frac{1}{\sqrt{2}} \begin{pmatrix} i & i \\ 1 & -1 \end{pmatrix}$ is unitary.

Example

Exercise. Unitarily diagonalise the matrix $A = \begin{pmatrix} 2 & -4 & 2 \\ -4 & 2 & -2 \\ 2 & -2 & -1 \end{pmatrix}$.

Solution. We can first observe that $A + 2I$ has nullity 2, so -2 is an eigenvalue with geometric (and hence algebraic, since A is self-adjoint) multiplicity 2. Since the sum of the eigenvalues is $\text{tr}(A) = 3$, the third eigenvalue must be 7.

We find $E_{-2} = \ker(A + 2I) = \ker \begin{pmatrix} 4 & -4 & 2 \\ -4 & 4 & -2 \\ 2 & -2 & 1 \end{pmatrix} = \ker \begin{pmatrix} 2 & -2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$,

and clearly $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \in E_{-2}$. We want to find another eigenvector in E_{-2} that is linearly independent with this one, but (ideally) also orthogonal to it.

Setting the first two components as 1 and -1 , we find $\begin{pmatrix} 1 \\ -1 \\ -4 \end{pmatrix} \in E_{-2}$.

Example

Exercise. Unitarily diagonalise the matrix $A = \begin{pmatrix} 2 & -4 & 2 \\ -4 & 2 & -2 \\ 2 & -2 & -1 \end{pmatrix}$.

Solution (continued). Finally, to find an eigenvector in E_7 orthogonal to the first two eigenvectors, since we are working over $\mathbb{R}^3$, we can easily construct one by taking the cross-product of the other two eigenvectors. So

it must be the case that $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} 1 \\ -1 \\ -4 \end{pmatrix} = \begin{pmatrix} -4 \\ 4 \\ -2 \end{pmatrix} \in E_7$.

Dividing the representative eigenvectors by their norms gives an orthonormal set which forms the columns of P . Thus $D = P^{-1}AP$ where

$D = \begin{pmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 7 \end{pmatrix}$ and $P = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{3\sqrt{2}} & -\frac{2}{3} \\ \frac{1}{\sqrt{2}} & -\frac{1}{3\sqrt{2}} & \frac{2}{3} \\ 0 & -\frac{4}{3\sqrt{2}} & -\frac{1}{3} \end{pmatrix}$ is unitary.

Spectral theorem for unitary operators

Corollary. (Spectral theorem for unitary operators)

Given an inner product space V over $\mathbb{R}$ or $\mathbb{C}$ with finite dimension n , and any **unitary** map $T \in \mathcal{L}(V, V)$ or matrix $A \in M_{n,n}(\mathbb{F})$, there exists an **orthonormal** basis for V made up of eigenvectors of T or A .

Proof. This again follows immediately from the fact any unitary operator is normal and the spectral theorem for normal operators.

Theorem. Given an inner product space V over $\mathbb{R}$ or $\mathbb{C}$ with finite dimension n , any **unitary** map $T \in \mathcal{L}(V, V)$ or matrix $A \in M_{n,n}(\mathbb{F})$ has all its eigenvalues of the form $e^{i\alpha}$ for corresponding $\alpha \in \mathbb{R}$.

Proof. Suppose $T(\mathbf{v}) = \lambda\mathbf{v}$ for some nonzero $\mathbf{v} \in V$ and $\lambda \in \mathbb{F}$. Then
 $\langle \mathbf{v}, \mathbf{v} \rangle = \langle T^*(T(\mathbf{v})), \mathbf{v} \rangle = \langle T(\mathbf{v}), T(\mathbf{v}) \rangle = \langle \lambda\mathbf{v}, \lambda\mathbf{v} \rangle = \lambda\bar{\lambda}\langle \mathbf{v}, \mathbf{v} \rangle = |\lambda|^2\langle \mathbf{v}, \mathbf{v} \rangle$.
Since $\mathbf{v} \neq \mathbf{0}$, this implies that $|\lambda|^2 = 1$, so λ lies on the unit circle, meaning it has the form $e^{i\alpha}$ for some $\alpha \in \mathbb{R}$.

Characteristic polynomial for unitary operators

Theorem. Given an inner product space V over $\mathbb{R}$ with finite dimension n , any unitary map $T \in \mathcal{L}(V, V)$ or matrix $A \in M_{n,n}(\mathbb{F})$ has characteristic polynomial

$$p_T(t) = (t-1)^a(t+1)^b \prod_{j=1}^k (t - e^{i\alpha_j})(t - e^{-i\alpha_j})$$

where $a + b + 2k = \dim(V)$ and $\alpha_j \in (0, \pi)$ for all j .

Proof. This follows immediately from the previous result and the fact that non-real roots of polynomials with real coefficients come in conjugate pairs.

Notice that though V is defined over $\mathbb{R}$, the characteristic polynomial for T might have non-real roots, in which case over $\mathbb{C}$ the eigenvalue $\lambda = e^{i\alpha}$ has a corresponding non-real eigenvector $\mathbf{v} = \mathbf{x} + i\mathbf{y}$ for $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. In that case, $T(\mathbf{v}) = \lambda\mathbf{v} = e^{i\alpha}(\mathbf{x} + i\mathbf{y}) = (\cos(\alpha)\mathbf{x} - \sin(\alpha)\mathbf{y}) + i(\sin(\alpha)\mathbf{x} + \cos(\alpha)\mathbf{y})$.

In particular, restricting T over the basis $B = \{\mathbf{x}, -\mathbf{y}\}$, we have

transformation matrix $[T]_B^B = \begin{pmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{pmatrix}$. This is a matrix representing rotation by α , also written as $R(\alpha)$.

Note in particular that $R(\alpha)$ is always orthogonal.

Transformation matrix for unitary operators

Corollary. Given an inner product space V over $\mathbb{R}$ with finite dimension n , for any **unitary** map $T \in \mathcal{L}(V, V)$ there is an orthonormal basis of V for which the transformation matrix of T is of the form

$$I_a \oplus (-I_b) \oplus R(\alpha_1) \oplus \cdots \oplus R(\alpha_k)$$

where $R(\alpha) = \begin{pmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{pmatrix}$, $a + b + 2k = \dim(V)$, and $\alpha_j \in (0, \pi)$ for all j .

Example. Consider the matrix $A = \frac{1}{3} \begin{pmatrix} 1 & 2 & -2 \\ -2 & 2 & 1 \\ 2 & 1 & 2 \end{pmatrix}$. We can check that

it is unitary (orthogonal), since each column vector has norm 1, and the dot product of any two distinct columns is 0. So the eigenvalues of A all lie on the unit circle, and have a product equal to $\det(A) = 1$, so at least one real eigenvalue of A is 1.

The matrix similar to A with form as described in the corollary must have trace $1 + 2\cos(\alpha)$ for some $\alpha \in \mathbb{R}$, and this must equal $\text{tr}(A) = \frac{5}{3}$. So $\cos(\alpha) = \frac{1}{3}$ and the matrix is $I_1 \oplus R(\arccos(\frac{1}{3}))$.